

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/1

PAPER 1 November 2024

1 hour

PAPER 1 — MULTIPLE CHOICE — 40 QUESTIONS — 1 HOUR

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2024 5006/1.

Additional materials: Answer all 40 questions. For each question there are four possible answers A, B, C and D.

INSTRUCTIONS: Choose the one you consider correct and record it. Each question carries 1 mark.

1. An acidic solution has **[1]**

- A) pH less than 7
- B) pH 14 only
- C) pH 7
- D) pH greater than 7

2. A dull black surface, compared with a shiny silver surface, **[1]**

- A) reflects all heat always
- B) is a good absorber and emitter of radiation
- C) cannot emit radiation
- D) is a poor absorber of radiation

3. Aerobic respiration in humans is summarised by **[1]**

- A) glucose + oxygen → carbon dioxide + water + energy
- B) glucose → lactic acid only
- C) carbon dioxide + water → glucose + oxygen
- D) protein + oxygen → urea

4. A plant cell in a concentrated sugar solution may become **[1]**

- A) plasmolysed
- B) a gamete
- C) photosynthetic
- D) turgid

5. Pressure is defined as **[1]**

- A) $F + A$
- B) F / A
- C) $F \times A$
- D) A / F

6. The main function of red blood cells is to **[1]**

- A) fight all pathogens by phagocytosis only
- B) make urea
- C) clot blood
- D) transport oxygen

7. Blue litmus paper in an acid [1]

- A) stays blue
- B) turns green
- C) turns red
- D) bleaches white always

8. During electrolysis, cations move to the [1]

- A) voltmeter
- B) anode
- C) cathode
- D) fuse

9. A vaccine works by [1]

- A) stimulating the immune system to make memory cells / antibodies
- B) killing all white cells
- C) replacing red cells
- D) neutralising stomach acid

10. The SI unit of speed is [1]

- A) kg
- B) m/s
- C) J
- D) N

11. The SI unit of force is the [1]

- A) newton
- B) kilogram
- C) watt
- D) joule

12. Which is a correct order in treating river water for drinking? [1]

- A) screening → filtration → chlorination
- B) chlorination → screening
- C) boiling → mining
- D) electrolysis → rusting

13. Temperature is measured with a [1]

- A) newton meter
- B) voltmeter
- C) thermometer
- D) ruler

14. A unit of density is [1]

- A) J
- B) N
- C) W
- D) g/cm³

15. The law of reflection states that [1]

- A) a mirror absorbs all light
- B) angle of incidence = angle of reflection
- C) angle of incidence = angle of refraction always
- D) light always bends towards the normal in air

16. An object has mass 40 g and volume 20 cm³. Its density is **[1]**

- A) 2 g/cm³
- B) 60 g/cm³
- C) 20 g/cm³
- D) 800 g/cm³

17. The SI unit of pressure is the **[1]**

- A) joule
- B) newton
- C) watt
- D) pascal

18. The arrows in a food chain show **[1]**

- A) the direction the animal walks
- B) the age of the organism
- C) the direction of energy flow
- D) wind direction

19. HIV is a **[1]**

- A) bacterium
- B) deficiency of vitamin C
- C) virus that attacks white blood cells / the immune system
- D) protozoan

20. Which is correct? **[1]**

- A) silver displaces sodium
- B) gold displaces zinc
- C) magnesium displaces copper from CuSO₄(aq)
- D) copper displaces magnesium from MgSO₄(aq)

21. In a food chain, a producer is an organism that **[1]**

- A) eats animals only
- B) makes its own food by photosynthesis
- C) breaks down dead matter only
- D) is always a fungus

22. Which statement is correct? **[1]**

- A) Conduction is bulk movement of a fluid
- B) Radiation does not need a medium
- C) Convection occurs in solids only
- D) Conduction occurs only in gases

23. Metals are good thermal conductors because they have **[1]**

- A) free electrons that transfer energy
- B) only covalent bonds
- C) no particles
- D) a vacuum inside

24. Urea is made in the **[1]**

- A) kidney from urine
- B) liver from excess amino acids
- C) skin from sweat only
- D) lungs from oxygen

25. The products of digestion are absorbed mainly in the **[1]**

- A) oesophagus
- B) bladder
- C) ileum (small intestine)
- D) trachea

26. A condom prevents pregnancy mainly by **[1]**

- A) increasing FSH always
- B) stimulating sperm production
- C) implantation of the embryo only
- D) preventing sperm meeting an egg

27. Each enzyme works best at its **[1]**

- A) optimum pH and temperature
- B) ignition temperature
- C) freezing point
- D) boiling point

28. A metal kettle is safer if it has **[1]**

- A) a higher current always
- B) no insulation
- C) a thicker live wire only
- D) an earth wire and a fuse / circuit breaker

29. A gas has **[1]**

- A) no fixed shape or volume
- B) fixed shape and volume
- C) fixed volume, no fixed shape
- D) particles that cannot move

30. Plasma transports **[1]**

- A) only cellulose
- B) only starch grains
- C) dissolved substances such as carbon dioxide, urea and hormones
- D) only oxygen in haemoglobin

31. The most abundant gas in clean dry air is **[1]**

- A) carbon dioxide
- B) argon
- C) nitrogen
- D) oxygen

32. The particle with a positive charge in the nucleus is the **[1]**

- A) proton
- B) electron
- C) neutron
- D) molecule

33. A cellulose cell wall is found in **[1]**

- A) viruses only
- B) plant cells only
- C) animal cells only
- D) both animal and plant cells

34. In a series circuit [1]

- A) resistance always decreases
- B) voltage is the same across each component always
- C) current is different in each component
- D) current is the same through each component

35. For main-group elements, the group number indicates the [1]

- A) number of outer-shell electrons
- B) number of electron shells
- C) mass number
- D) number of neutrons

36. Universal indicator in a strong alkali is [1]

- A) purple / blue
- B) green
- C) red
- D) colourless always

37. Electrolysis requires [1]

- A) pure water with no ions, always
- B) a sugar solution
- C) a molten or aqueous ionic compound
- D) a solid ionic compound with no mobile ions

38. Electrons are found in [1]

- A) neutrons only
- B) the nucleus
- C) electron shells / energy levels
- D) the vacuole

39. Aerobic respiration takes place mainly in the [1]

- A) mitochondrion
- B) chloroplast
- C) cell wall
- D) nucleus

40. Acid + carbonate produces [1]

- A) salt + oxygen
- B) water only
- C) salt + hydrogen
- D) salt + water + carbon dioxide

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).